Aaryan Wadhawan

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Education

VIT Bhopal University **2022 – June 2026**
B.Tech in Computer Science (Cyber Security and Digital Forensics) | CGPA: 8.15 / 10.0 *Kothrikalan, MP*

Projects

ARistotle — AI-Powered AR Object Explainer | GitHub

- Built a browser-native AR object explainer using **YOLOv8s** for real-time object detection on live webcam frames every 3.5s; detected labels passed to **LLaMA 3 via LangChain** to generate explanations rendered as floating AR cards using AR.js + A-Frame — no app install required.
- Implemented **in-memory LLM response caching** that reduced Groq API calls by ∼90% per session, keeping inference within free-tier limits at sub-500ms response time.
- Resolved a silent OOM crash on Render by switching to **CPU-only PyTorch wheels**, eliminating CUDA bindings loading on a GPU-less server — deployed live with FastAPI backend and Netlify frontend.

DocuMind — RAG Document Q&A System | GitHub | Live Demo

- Built an end-to-end **RAG pipeline** using FastAPI, LangChain, FAISS, and LLaMA 3 (Groq API); users upload PDFs and receive context-grounded answers with source citations at **sub-2-second latency**.
- Engineered document ingestion with **500-token chunking** and semantic vector indexing; retrieves top-3 relevant chunks via cosine similarity before passing context to the LLM.
- Containerised full stack with **Docker Compose** (5 services); **GitHub Actions CI/CD** runs pytest on every push and builds Docker image only on passing tests.
- Integrated **MLflow** experiment tracking and **Prometheus + Grafana** monitoring; logs every inference run with latency and model metadata for performance comparison.

ML-Based DDoS Detection and Mitigation on SDN | GitHub

- Trained Decision Tree, Random Forest, and ANN on **CIC-IDS 2017** network traffic; achieved **92.6% accuracy** with Random Forest after feature engineering, normalisation, and class-imbalance handling.
- Integrated real-time mitigation logic into SDN controllers to dynamically block detected attack flows, reducing simulated manual threat-response time.

Technical Skills

Languages: Python, C, C++, JavaScript, TypeScript
Machine Learning: TensorFlow, Keras, scikit-learn, Neural Networks, Feature Engineering
AI Engineering: LangChain, RAG, LLMs, FAISS, HuggingFace Inference API
Computer Vision: YOLOv8, OpenCV, AR.js, A-Frame
MLOps & DevOps: Docker, Docker Compose, Kubernetes, GitHub Actions, Prometheus, Grafana, MLflow
Backend & Web: FastAPI, REST APIs, React.js, Streamlit, HTML, CSS, Tailwind CSS
Databases: MySQL, MongoDB, Firebase
Cloud & Tools: AWS, Render, Netlify, Git, GitHub

Achievements

- Ranked in **Top 50 teams** out of all submissions at Visionary Hackathon 2.0 (Metti Xathon) GitHub
- Solved **250+ DSA problems** across LeetCode, GeeksforGeeks, and Codeforces.

Certifications

- Oracle Generative AI Professional
- IBM DevOps, Agile and Design Thinking